Reflexes of Proto-Algonquian initial *t in Blackfoot

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Contents

1 Overview

Blackfoot has long been labelled "divergent" with respect to the rest of the Algonquian family because of its lexical differences (Voegelin 1941: 506) and phonological innovations (Goddard 1994b: 187).[1] Even fairly recent studies of Blackfoot historical phonology propose a number of unusual sound changes (Berman 2006; Proulx 1989). This paper focuses on one such sound change in by Berman (2006), who argued that Proto-Algonquian (PA) word-initial *t becomes *k* in Blackfoot on the basis of three reconstructions.

This paper uses multiple methodologies to examine these three reconstructions in detail, including comparative evidence with other Algonquian languages, historical evidence from older Blackfoot resources, synchronic morphemic analysis, and explicit derivations of each form. Ultimately, I find that one of the three reconstructions is supported by all available methodologies, one is ruled out by the relative chronology of known sound changes in Blackfoot, and one apparently has two reflexes in Blackfoot which are in complementary distribution but which involve spurious sound changes. Still, the fact that one reflex requires a major place change suggests that this process may have also occurred in other forms in Blackfoot. Future work should investigate this hypothesis further.

The paper is organized as follows. In Section 3 I explain the different methods I used to investigate these three reconstructions, as well as other conventions like abbreviations and transcriptions. Section 4 includes necessary phonological background, including aspects of the phonology of Proto-Algonquian and Blackfoot, and relevant known historical changes in Blackfoot. The main data and sound changes are discussed in Section 5, and in Section 6 I conclude.

[1]The words in this paper and the knowledge they impart belongs to the Blackfoot people. I am indebted to several formative conversations about Algonquian historical phonology with Richard Rhodes, Ives Goddard, David Pentland, Rose-Marie Déchaine, Donald Frantz, Inge Genee, Conor Quinn, Monica Macaulay, Joe Salmons, Dave Costa, Hunter Lockwood, and other members of the University of Wisconsin-Madison "Algonquian Reading Group" in 2024. Any mistakes here are surely my own.

2 Background

Berman (2006: 265) argues that PA initial *t shifts to B *k*. This sound change requires a "major place change" from dorsal to coronal, which is relatively rare outside of Austronesian (Blevins 2004; Blust 2004). Blackfoot thus stands to have an impact on theories of historical change and phonology if the sound changes can be robustly confirmed. Furthermore, Berman (2006) bases this claim on just three pairs of Blackfoot and Proto-Algonquian forms, raising the question of how robust the reconstructions are for Blackfoot. In this section I briefly review Berman's proposal.

Berman (2006) points out that initial *t* is rare in Blackfoot, and proposes the change of PA *t > B *k* to account for this gap. The rarity of initial [t] in Blackfoot can be seen by counting the number of entries under each section of the Frantz & Russell (2017) dictionary. An R script was used to extract entries from the original Word document underlying the dictionary. These were placed into a text document with tab separated values which was then double-checked for accuracy and counted. The corpus contains 5,180 entries in total, including some entries which appear in the Word document but not in the published dictionary. The counts of entries in each dictionary section are shown in Table 1. The 29 entries listed under <t> are substantially lower than the number of entries listed under the other consonants, which typically range between 150 and 250.[2] According to Berman, Blackfoot initial *t* "does not appear to be the regular outcome of any PA consonant" (Berman 2006: 275).

Table 1: Number of entries listed under each section in Frantz & Russell (2017)

Section	Vowels	h	k	m	n	p	s	t	w	y
Total = 5,180	3,025	8	211	267	167	142	816	29	314	201

[2] The counts under <h> are small because words cannot begin in a phonemic /h/ in Blackfoot. The eight words listed under <h> are all uninflected interjections, a category for which initial [h] is predictable (Taylor 1969: 30). The counts under <s> are unusually high because of the unusually wide distribution of *s* (see Weber & Derrick 2024): this entry contains stems beginning with short *s* and long *ss* before vowels and consonants.

The three pairs of PA reconstructions to Blackfoot forms in (1) were used to support Berman (2006)'s (2006) proposal. I have altered Berman's PA transcriptions to match the conventions in Goddard (1994b), as explained in Section 3.4.

(1) BLACKFOOT REFLEXES OF INITIAL PA *t (Berman 2006: 275)

Proto-Algonquian		Blackfoot		Rec. No.
***t**epeskatwi	'it is night'	**k**o'kó-	II 'be night'	#66
***t**epeskwi	'night'	**k**o'kóyi	'last night'	#66
***t**aw-	'open'	-i**k**aw-	adt. 'open'	#65

The first and second examples are clearly related: the first contains a verbal II stem *tepesk-at- 'be night', while the second contains a nominal NI stem *tepeskw- 'night'. The third example is an "adjunct" in Frantz & Russell (2017), which is a root which can be used as an initial or preverb. The question I ask in this paper is whether the Blackfoot forms are likely reflexes of the Proto-Algonquian reconstructions, using the methods discussed in the next section.

3 Methods and conventions

I begin by describing the set of methods used to investigate Berman's three reconstructions (Section 3.1). I then list the sources used for each language and my workflow for compiling cognate sets (Section 3.2). I end by explaining the abbreviations in the paper (Section 3.3) and the format of each reconstruction (Section 3.4).

3.1 Multiple methodologies

The main methodology used in this paper is the comparative method (e.g. Fox 1995). I additionally consider older historical records of Blackfoot, the synchronic morphology of

Blackfoot, and the relative chronology of known sound changes. Though the comparative method was instrumental in determining PA reconstructions (e.g. Bloomfield 1946; Siebert Jr. 1941), it has not be used to investigate sound changes in Blackfoot. Previous papers on Blackfoot historical phonology instead compare abstract Blackfoot forms (frequently a root or stem) to abstract Proto-Algonquian reconstructions (Berman 2006, 2007; Proulx 1989, 2005; Taylor 1960, 1967; Thomson 1978).

Working off of reconstructed components and an idealized gloss can be misleading, because the researcher is unable to see the full range of semantics associated with each root in Blackfoot or across the family. Because each root is examined in isolation, it is easy to posit spurious sound changes to account for the root in question. To address this, I compile cognate sets by finding reflexes across nine languages, including Blackfoot, which span all three major areal groupings in Algonquian. I include full words (or sometimes stems) in the cognate sets, so that the reconstructable portion (often a root or stem) is shown across multiple phonological contexts and semantic frames. For the case of PA *taw- ‘hole, opening’ below, this method helped me find a second reflex in Blackfoot which overlapped in semantics with reflexes in other languages. The comparative method also shows that both putative reflexes in Blackfoot for PA *taw- ‘hold, opening’ involve irregular sound changes.

Blackfoot has undergone many innovative sound changes, and it is likely that some of these occurred recently enough that they are reflected in the historical record. The Blackfoot Words database (Weber, Brown, et al. 2023) is a lexical database of words digitized from multiple sources. Tokens of the same stem or morpheme across multiple sources and orthographies are linked to standardized lemmas. Although diachronic change is not directly captured in the database, viewing all the tokens linked to a lemma provides a way for researchers to find archaic spellings. I checked the digitized sources in Blackfoot Words for archaic spellings. This was especially useful for finding older spellings of words related to ‘night’, some of which still include a coronal reflex of *t in medial positions.

Because often only a root can be reconstructed for Blackfoot, it helps to understand the synchronic morphology of stems. The dictionary, Frantz & Russell (2017), often includes separate entries for "adjuncts", which are roots which can be used as preverbs or initials. However, there are a number of roots which only occur in initial position, and these are only found as part of stem entries. Understanding the internal composition of stems helped to find the second putative reflexes of PA *taw- 'hole, opening'. Future versions of the Blackfoot Words database will also include morphological analysis, which should help support projects of this kind.

Even when previous papers posited Blackfoot sound changes, they did not always consider the relative chronology of these sound changes. In Section 4 I summarize the evidence for the sound changes which occur in all three reconstructions in this paper. When the relative chronology can be reliably determined, I also explain how these rules are ordered with respect to one another. For each of the three reconstructions, I include a derivations from the PA reconstruction to the modern reflex. Sometimes the relevant order of sound changes rules out a putative reconstruction. This is the case for PA *tepeskwi NI 'night', which I argue cannot be the reconstruction for B *ko'kóyi* 'last night'.

3.2 Sources and workflow

The eight languages in this paper represent each of the three areal groupings in Algonquian (Mithun 1999). Only one of these three (Eastern Algonquian) has been established as a genetic subgroup (Goddard 1974, 1980), though even this has been contested (Pentland 1992; Pentland 1979; Proulx 1984).

(2) Distribution of languages across Algonquian areal and genetic groupings

a. Plains Algonquian: Blackfoot, Arapaho, Cheyenne

b. Central Algonquian: Plains Cree, Menominee, Ojibwe, Meskwaki, Shawnee

c. Eastern Algonquian: Passamaquoddy

Unless otherwise noted, the following sources were used.

- Arapaho: Kazeminejad, Cowell, & Hulden (2017); cited with a unique id like [L12345].
- Cheyenne: Fisher et al. (2006)
- Plains Cree: Wolvengrey (2001) (digitized by Arppe et al. 2022)
- Ojibwe: Livesay & Nichols (2024)
- Shawnee: Selstad (1998)
- Passamaquoddy: Language Keepers & Peskotomuhkati-Wolastoqey Dictionary Project (2024)

Traditionally, Proto-Algonquian reconstructions have used the entire inflected word (e.g. Bloomfield 1946). This is not feasible for Blackfoot, because the reflexes of initials and finals have frequently been rearranged into new stems, such that "few complete words match those of other members of the family" (Pentland 2023: *ix*).[3] Because of this, I reconstruct the largest constituent that I am able.

There are now a number of digital tools and lists of components to help researchers find likely cognates. As a first step, I found Proto-Algonquian components that looked plausibly similar to Blackfoot roots in several of these resources (Hewson 2024a; Macaulay, Lockwood, & Hieber 2024). I then checked Proto-Algonquian dictionaries for reconstructions that looked related to Blackfoot forms and added those to my working list (e.g. Aubin 1975; Hewson 1993; Pentland 1979). Only after compiling a list of reconstructed components and words did I turn to dictionaries of modern languages, where I found cognates for each component in a variety of phonological contexts with a representative

[3] Although this problem is acute for Blackfoot, this kind of recombination occurs to some extent in all of the languages. Some recent publications "reconstruct" a Proto-Algonquian form even if it only exists in a single language, as long as the components are known from other cognates (e.g. Hewson 2024a,b; Pentland 2023). Other databases focus on comparing the components themselves across languages (e.g. Macaulay, Lockwood, & Hieber 2024).

range of meanings. Finally, I searched older Blackfoot sources within Blackfoot Words (Weber 2022; Weber, Brown, et al. 2023) to find additional instances of the forms in other phonological or morphological environments.

3.3 Abbreviations

Sources

[A #]	Aubin (1975) (# = reconstruction)
[FR #]	Frantz & Russell (2017) (# = page)
[H #]	Hewson (1993) (# = reconstruction)
[P #]	Pentland (2023) (# = page)

Languages

PA	Proto-Algonquian	M	Menominee
B	Blackfoot	O	Ojibwe
Ar	Arapaho	Mk	Meskwaki
Ch	Cheyenne	Sh	Shawnee
C	Plains Cree	Ps	Passamaquoddy

Other symbols

*	reconstructed proto-form
†	uncertain reconstruction
$>$	changed to
$<$	changed from
$\rightarrow$	morphological replacement

3.4 Formatting

Each reconstruction begins with a numbered header in bold, as below. This header gives my PA reconstruction and the Blackfoot reflex. My Proto-Algonquian transcriptions follow Goddard (1994b). The header is preceded by a dagger (†) if I am not fully confident about the reconstruction, either because it is onomatopoeic or because there are unusual sound changes. I indicate whether the reconstruction is an initial, stem, or word in square brackets to the right. For the reconstructions and reflexes, I use a hyphen between stem internal components and between suffixes. If only part of a word is cognate, I underline the cognate part.

2. PA *tepesk-at-wi 'it is night' > B *ko'ko-wa* 'it is night' [word]

Below the header I give the cognate set, as below. The first lines always list previously proposed PA reconstructions, with their original transcriptions. This is followed by one or more Blackfoot forms. If there are few forms in Frantz & Russell (2017), I simply list them all so that there is a good representation of the semantic interpretation of the initial or stem. Otherwise, I choose a subset to represent the reconstructed form in a variety of phonological environments with a variety of meanings.

PA	*tepexk-at-	'be night' (Goddard 1982: 37, note 109)
	*tepexk-at-wi	II 'it is night' [P 3137]
B	kòoʔkó-ʔwa	it is night (Taylor 1969: 32)
	áká-aks-i<u>ko'ko-wa</u>	it will be night soon [FR 57]

The remaining languages are presented in a fixed order: Arapaho (Ar), Cheyenne (Ch), Cree (C), Menominee (M), Ojibwe (O), Meskwaki (Mk), Shawnee (Sh), Passamaquoddy (Ps). Thus, the Plains languages are grouped at the top, the Central languages in the middle, and the Eastern languages (represented by Passamaquoddy) at the bottom.

4 Relevant sound changes and relative chronology

In this section I summarize the sound changes from Proto-Algonquian (PA) into Blackfoot which are relevant for the cognate sets in this paper. The sound system of PA is well-established via phonological reconstructions (cf. Aubin 1975; Bloomfield 1946; Goddard 1979; Hewson 1993; Pentland 1979, 2023; Siebert Jr. 1941). Previous papers on Blackfoot historical phonology include (Berman 2006, 2007; Proulx 1989, 2005; Taylor 1960, 1967; Thomson 1978), with Berman (2006) and Proulx (1989) being the most comprehensive of this set. The reconstructions and Blackfoot forms in this section are verbatim from Berman (2006), except that I have altered his PA transcriptions to match Goddard (1994b) and also included morpheme breaks and underlining. Previous research (including Berman 2006) does not always make the rules explicit nor always discuss rule ordering, so I do that here as well.

In terms of derivational morphology, the abstract II finals *-at, *-an, and *-en are replaced by Blackfoot *-o*, as in (3).

(3) Abstract II final replacement (Berman 2006: 266; Proulx 1989: 48)

*ki·šek-**at**-wi	'it is day'	-iksistsik-**ó**-	II 'be day'
*kwesekw-**an**-wi	'it is heavy'	-(i)ssok-**o**-	II 'be heavy (weight)'
*ni·p-**en**-wi	'it is summer'	niip-**ó**-	II 'be summer'.

Berman (2006: 281) suggests that this is due to regular sound changes. First, PA *n and *t are both lost before the word-final II suffix *-wi. Second, *a and *e round before *w, as they do in other contexts. A sound change would explain why the same change affects an AI stem, PA *kwečpan 'fear' (Pentland 1983: 383) > B *kooʔpu-* 'fear (AI)' (cited in Proulx 1989: 48). Proulx (1989: 48) instead suggests that the Blackfoot *-o* is an extension of the AI *-o 'middle reflexive' (i.e., 'patient descriptive'; Denny 1984: 246). I find this to be a less likely source of Blackfoot *-o* because it only affects this stem or very few stems. The regular pattern is to equate abstract II finals with *-o*.

The rule below is written simply as a morphological replacement rule, abstracting away from Berman's (2006) phonological analysis.

R1 II final replacement: {*-at, *-an, *-en} → -o

PA short *e rounds to B *o* if followed by an *o or *o· in the next syllable (Berman 2006: 266, 279–280), as in 'it snows', (4). According to Berman (2006: 279–280), the triggering *o or *o· may be original or derived by earlier sound change. Rounding also occurs before a following *kw (Berman 2006: 266), as in 'head', (4).

(4) Short *e rounding (Berman 2006: 268, 280; Proulx 1989: 55)

*m**e**spo-wi	'it snows'	-**o**hpotaa-	II 'snow'
*me-št**e**kwa·n-i	'head'	mo-'t**o**káán-i	NI 'head, hair'

Both rules of short *e rounding can occur across clusters. Short *e rounds before *o across a cluster in PA *m**e**spowi 'it snows', (4), but also in PA *m**e**htawaki 'ear' > B *m**o**htóókisi* 'ear' (Berman 2006: 282). Short *e rounds before *θkw in PA ***e**θkwe·wa 'woman' > B -***o**hkiimi*- AI 'have a wife' (Berman 2006: 281).

I have split short vowel rounding into two rules, R2a and R2b, based on the conditioning environment.

R2 Short vowel rounding:

R2a *e > *o / __(C)Co

R2b *e > *o / __(C)kw

The data in the next section show that R1 is crucially ordered before R2a.

PA *kw regularly becomes B *ksis* before a short vowel at the end of a word, as in (5).

(5) Stem-final PA *kw > B *ksis* (Berman 2006: 278, 282; Proulx 1989: 59)

*me**kw**-i	‘awl’	mo**ksis**-a	NA ‘awl’
*me-kete**kw**-i	‘knee’	mo-tto**ksís**-a	NA ‘knee’
*me-θen**kw**-i	‘armpit’	mo-’**ksis**-i	NI ‘armpit’

Note that R3 must follow R2b, because the *kw in ‘knee’ causes the preceding *e to round to *o* (via R2b) but *ksis* (the outcome of R3) is not expected to cause rounding.

R3 Final *kw fission and assibilation: *kw > ksis / __V#

In all other positions, post-consonantal *w deletes (Berman 2006: 268, 271; Proulx 1989: 55), as in (6). For example, the *kw in ‘snare’ becomes B *k*, and the *pw in ‘he bites him’ becomes B *p*.

(6) Post-consonantal *w deletion (Berman 2006: 268, 271; Proulx 1989: 55)

*na**kw**a·-	‘snare’	-o**k**aa-	AI ‘rope, snare’
*saki**pw**-e·-wa	‘he bites him’	-(i)siksi**p**-	TA ‘bite’

R4 must occur after R2a and R2b (short vowel rounding), so that *kw causes a preceding short vowel to round to *o* before *kw > *k*. R4 must also follow R3; otherwise, R4 would bleed R3.

R4 Post-consonantal *w deletion: *w > Ø / C__

There are numerous cases where a PA short vowel (often *e) deletes and has no reflex in Blackfoot, as in (7). Various aspects of Blackfoot vowel syncope are discussed in Thomson (1978), Proulx (1989: 47), & Berman (2006: 266, 2007).

(7) Short vowel syncope (Berman 2006: 267, 270, 282; Thomson 1978: 253)

*k**e**p-	‘cover up, block’	-ipp-ohsoyi-	AI ‘suffocate, smother’
*me-k**e**tekw-i	‘knee’	mo-ttoksís-a	NA ‘knee’
*aθ**o**sky-	‘work’	-o’k-oo-	AI ‘become exhausted, …’

According to Berman (2006: 266,267), the first vowel of a stem deletes if that vowel is short and the stem added an i-prefix to form a non-initial stem, as for PA *kep- ‘cover up, block’ > B *ipp*- in (7). The second vowel of the stem deletes if the first two vowels of the stem are both short, separated by a single consonant, and the second vowel is not *i, as for ‘knee’ and ‘work’ in (7). Although Berman claims that all short vowels syncopate in these positions, it typicaly affects short *e and *o. There is only one putative example of short *a syncopation (*pak- ‘pound, strike’ > -*ikk*- vrt. ‘hit, strike’; Berman 2006: 271). Later vowels may also syncopate, but the exact conditions are not fully understood. The only rule relevant for the data in this paper is for second vowel syncope, as in R5.

R5 Short vowel syncope: {*e, *a, *o} > Ø / #(C)VC__

When a vowel deletes via R5, two or more consonants are brought adjacent to one another. In general, if syncope causes two consonants to become adjacent, the first consonant assimilates completely to the second. If syncope causes three consonants to come together, the cluster resolves to a glottal stop followed by the third consonant. However, not all combinations of consonants have been studied in Blackfoot historical phonology, and it may be that other patterns are found in further reconstructions (see Weber 2016).

R6 Secondary cluster resolution

R6a Consonant assimilation: $C_1 + C_2 > C_2C_2$ / __V

R6b Cluster resolution: $C_1 + C_2 >$ ʔ / __C

Initial PA *e neutralizes with *i in Blackfoot. This was a very early rule in all of the western Algonquian languages (Pentland 1979: 403; Oxford 2015: 320–321; Goddard 1994b: 192–193).

R7 *e > *i / # __

Medial PA *t is resolved in one of several ways, and the conditioning environments for assibilation point to a particular rule ordering. PA *i, *i·, and also initial *e both cause a PA *t to pre-assibilate to *st* (Proulx 1989: 52; Berman 2006: 265; Michelson 1935: 142–143). Because initial PA *e is treated the same as inherited *i, R8 must be ordered after R7.

R8 *t > *st* / *i(·) __

When short vowels do not round or syncopate, they are resolved in the following ways (after Berman 2006: 266). Initial and final short PA *a remain B *a*, but otherwise all short vowels merge to B *i*. Because medial PA *a and *e do not cause a *t to preassibilate to *st*, R9 must be ordered after R8. It is possible that some of these mergers occurred earlier than others. For example, since PA *a and *e undergo syncope by R5, but short *i is excluded from syncope, it is possible that *a and *e merged at some earlier stage to an intermediate vowel (possibly *e) before later merging with *i.

R9 Basic reflexes of short vowels

- **R9a** *a > *a* / {#__, __#}
- **R9b** *a > *i*
- **R9c** *e > *i*
- **R9d** *i > *i*
- **R9e** *o > *o*

Finally, medial PA *t is resolved two ways (Berman 2006: 265; Proulx 1989: 50, 53; Michelson 1935: 142–143). It becomes B *st* after PA *i, *i·, or initial *e, B *ts* before an *i* from any historical source, and otherwise becomes B *t*. Because medial *a, *e, and *i all behave the same with respect to this rule, R10 must be ordered after R9.

R10 *t > *ts* / __ *i

With those ten rules in place, I now turn to a more detailed look at the three reconstructions from Berman (2006).

5 Data

In this section I walk through each of the three reconstructions from Berman (2006). I find strong support for a *t > *k* change in PA *tepesk-at-wi 'it is night' > B *ko'ko-wa* 'it is night' (Section 5.1). However, the other two reconstructions are not as strong. The relative chronology established in the previous section rules out PA *tepeskw-i 'night' > B *ko'ko-wa* 'it is night' (Section 5.2). Finally, there are two plausible reflexes of PA *taw- 'open' (Section 5.3). Although these occur in complementary distribution, suggesting a single source, neither involves regular sound changes.

5.1 Supported: *tepesk-at-wi 'it is night' > *ko'ko-wa* 'it is night'

The reconstruction of PA *tepesk-at-wi 'it is night' > B *ko'ko-wa* 'it is night' is the best supported of the three reconstructions from Berman (2006). There is abundant comparative evidence to support this reconstruction, and the sound changes from PA to Blackfoot are regular and follow the relative chronology argued for in the preceding section. Further support comes from some archaic forms in Blackfoot with this stem occurring in word-medial position after a root.

There are reflexes of this form in multiple Central Algonquian languages (Menominee, Ojibwe, and Meskwaki) as well as Eastern Algonquian (here represented only by Passamaquoddy). In Menominee and Passamaquoddy, I only found examples of this stem after a root, suggesting it has been reanalyzed as a "deverbal" (bound) stem (see Goddard 1990: for the terminology).

1. PA *tepesk-at-wi 'it is night' > B *ko'ko-wa* 'it is night' **[word]**

PA	*tepexkat-	'be night' (Goddard 1982: 37, note 109)
	*tepexkatwi	II 'it is night' [P 3137]
B	kòoʔkó-ʔwa	it is night (Taylor 1969: 32)
	áká-aks-iko'ko-wa	it will be night soon [FR 57]
	áka-iko'ko-wa	it is night [FR 57]
Ar		
Ch		
C		
M	ahpeˑhta-tɛpɛhkat	II 'it is so far into the night, so late at night' (Bloomfield 1975: 12)
	kɛqsi-tepɛˑhkat	II 'it is a cold night' (Bloomfield 1975: 77)
	keˑsawe-tɛpɛˑhkat	II 'it is a warm night' (Bloomfield 1975: 90)
	kiˑskaniˑ-tepɛˑhkat	II 'it is dark; it is night' (Bloomfield 1975: 98)
O	dibikad	vii it is night; it is dark (as at night)
	aabitaa-dibikad	vii it is midnight
	biichi-dibikad	vii it is a long night
Mk	tepehkatwi	'it is night' [P 3137], who cites Si75 [Mesk.]
Sh		
Ps	pomi-tpuhkot	the night goes along
	tki-tpuhkot	it is cold night

The Blackfoot reflex can be derived with regular sound changes, as in **??**. I have reminded the reader of each rule that applies, as well as the crucial orders established in the previous section. First, the short *e in the second syllable deletes by R5, and the resulting consonant cluster of *p + sk resolves to *ʔk by R6b. The VII suffix *-at is replaced by *-o via R1. Regressive rounding of *e > *o is then able to occur by R2a. Note that R1 and R2 are not crucially ordered with R5 and R6b, and it is possible that the latter two occurred earlier than the first two rather than later. No matter the order, once all the rules have taken place, the initial *t backs to *k. The Blackfoot -*wa* '3' suffix is innovated, as discussed in Goddard (2018).

Table 4: Derivation of B *ko'ko-wa* by rule

PA	*tepesk-at-wi 'it is night'		Crucial ordering
1.	tepsk-at-wi	R5. *e > Ø	(R5 > R6b)
2.	teʔk-at-wi	R6b. *ps > *ʔ / __C	
3.	teʔko-	R1. *-at → *-o	(R1 > R2a)
4.	toʔko-	R2a. *e > *o / __(C)Co	
5.	koʔko-	*t > *k / #__o(ʔ)k	
B	ko'ko-wa 'it is night'		

Some archaic forms and older documentation support the idea that B *ko'ko-* 'night' once began in *t. Specifically, some archaic non-initial noun stems begin in B *st,* which is the regular reflex of medial PA *t after *i. The words for 'midnight' and 'clear night' below are composed of prenoun *tatsik-* 'middle' or *sopowa-* 'clear' plus the noun stem 'night'. (See the next section for arguments that Blackfoot's noun stem is also a reflex of the verbal II stem *tepesk-at- II 'be night', rather than a reflex of the noun stem *tepeskw- 'night'.) These noun stems were recorded with an initial *st* instead of a *k* by in Uhlenbeck (1938) and Holterman (1996), both resources which record the Blackfeet (Southern Piegan) dialect. The same form for 'midnight' was marked as archaic in Frantz & Russell (2017: 135).

(8)	katsíks-i**st**ó'ko-yi	midnight (in the middle of the night) (archaic) [FR 135]
	tátsiks-i**st**oku-i	midnight (Uhlenbeck 1938: 66)
	tazikx-i**st**oko-i	midnight (Holterman 1996: 31)
	sipowa-i**st**oko	clear night (Holterman 1996: 14)

I take this as additional evidence that the Blackfoot stem derives from a PA stem in *t. The reason is that even if a rule changed initial PA *t > B *k*, the reflex of medial *t is expected to be *st* after *i. This suggests that the Blackfeet dialect at this time had two forms for the stem 'night': it would be *ko'ko-* in initial position but -*(i)sto'ko* after a root. Other dialects may have already leveled out this alternation and begun to use *ko'ko-* in all positions. For example, Tims (1889) records the Siksiká dialect decades before Uhlenbeck (1938) recorded the Blackfeet dialect. He also records an instance of 'midnight', but it includes a *k* rather than *st*: *itŭt'siki-aikoko* 'midnight' (Tims 1889: 148).

5.2 Contradicted: *tepeskw-i 'night' > *ko'ko-wa* 'it is night'

Berman (2006: 275) argues that the Blackfoot noun *ko'kóyi* 'last night' is a reflex of PA *tepeskwi 'night' (Goddard 1982: 37, note 109) with a stem-final *w. Although this form looks related to other reflexes, as in the cognate set below, the relative chronology of sound changes proves that the Blackfoot noun is not a reflex of PA *tepeskw- NI 'night'.

2. PA *tepeskw-i NI 'night' [word > stem]

PA	*tepexkw-i	NI 'night' (Pentland 2023: 3139)
PA	*tepexkw-i	(Goddard 1982: 37, n. 109)
B		
Ar	téce'	/tece'/ ni night [L21883]; (Goddard 1982: 36–37)
Ch	taa?e	pl. *taa?ėstse* (Goddard 1982: 36–37)
	É-taa'e-ve	It is night.
C	tipisk	night
M		
O	tipikk	'night' [P 3139]
Mk		

Sh

Ps

Recall that in reconstruction like PA *me-ketekw-i ‘knee’ > B *mo-ttoksis-a,* the stem-final *kw causes a preceding *e to round to *o (R2b) before stem-final *kw becomes *ksis* (R3). This rule order predicts that the reflex of PA *tepeskwi NI ‘night’ in Blackfoot should be *ko’ksisi* ‘night’, as in Table 6. Since the Blackfoot noun stem in *ko’kó-yi* ‘last night’ looks exactly like the verb stem in *ko’kó-wa* ‘it is night’, some other explanation is needed.

Table 6: Incorrect derivation of B *ko’ko-wa* by rule

PA	*tepeskw-i NI ‘night’		Crucial ordering
1.	tepskw-i	R5. *e > Ø	(R5 > R6b)
2.	teʔkw-i	R6b. *ps > *ʔ / __C	
3.	toʔkw-i	R2b. *e > *o / __(C)kw	(R2b > R3)
4.	toʔksis-i	R3. *kw > *ksis / __V#	
B	*ko’ksis-i ‘night’		

Proulx (1989: 48–49) notices the same rule ordering issue for the Blackfoot noun *ksi-istsikó-yi* ‘day’, which does not look like a regular reflex of PA *ki·šekw-i ‘day’, because it does not end in expected *ksis-i.* He suggests that the nominal stem has been replaced with the II verb stem, and the only difference between the two synchronically is the inflectional ending. I argue that the same thing must have occurred here, and that the stem in Blackfoot noun *ko’kó-yi* is actually a reflex of the PA verb stem *tepeskat- II ‘be night’.

Proulx’s observation is quite general in the language: Blackfoot stative verbs can be productively used as nouns. For example, the II stem *si’tsii-* ‘smoke’ in *iisí’tsii-wa kitótotan-nooni* ‘our fuel smoked’ (Frantz & Russell 2017: 256) and *sayíiʔtsii-ʔwa* ‘it smoked (fire)’ (with initial change) (Taylor 1969: 122) can be used as an inanimate noun *siiʔtsíí* ‘smoke’ (Taylor 1969: 122). Similarly, the AI stem *sikimi-* ‘be black (of animals) in *síkimi-ʔwa* ‘he is black’ (Taylor 1969: 75) can be used as an animate noun *síkimi* ‘black horse!’ (Voc)

(Curtis 1911: 36). Regarding animate nouns, **Bliss:2013phd** argues that the suffix *-wa* is neutral with respect to category and can be used on nominal and verbal stems. However, it seems this process extends beyond animate nouns to all stative nouns, regardless of animacy.

5.3 Multiple reflexes: †*taw- 'hole, opening'

The final reconstruction from Berman (2006: 275) with a PA *t > B *k* reflex is PA *taw- 'open' > B *-ikaw-* adt. 'open'. He cites the non-initial stem, which has an "*i*-prefix" (Berman 2006: 267). Because medial short *a becomes B *i* by R9b, he proposes that *-ikaw-* actually contains a reflex of PA *a· > B *aa*, which shortens regularly after the *i*-prefix. Synchronic alternations support the idea that the vowel is long in Blackfoot. It is long at the left edge and short medially after the *i*-prefix: *kaaw-* ∼ *-ikaw-*.

(9) *<u>kaaw</u>áí'piksit!*
kaaw–ai'piksi–t
open–pull.TI–2SG.IMP
'open it!' [FR 44]

(10) *áaksi<u>kaw</u>ai'piksima*
aak–kaaw–ai'piksi–m–a
FUT–open–pull.TI–IND–3
'she will open it' [FR 44]

A closer look reveals that the root PA *taw- 'hole, opening' has two probable reflexes in Blackfoot: *ao-* 'hole, opening' before consonants and *kaaw-* 'open' before vowels and nucleic *ss*. This is shown in the cognate set below. (Note: the straight apostrophe in Tims means that the preceding syllable is stressed.) I have given a number of examples from each language to demonstrate the full range of semantics, which includes an opening, a gap, or space created by humans or nature. These semantics overlap with the range of semantics of *ao-* and *kaaw-* in Blackfoot. The Cheyenne initial is longer than expected, as if from *tawa·Ck-, where C is a non-nasal consonant. Regardless of the source of the extra material, the first part of the initial is certainly cognate. In Passamaquoddy the typical reflex of PA *a is Ps *o*, which represents a schwa. This schwa is often rounded to *u* before *w* (LeSourd 1993: 70), giving *tuw-* in the cognate set below.

3. †PA *taw- ‘hole, opening’ > B *ao-*, *kaaw-* ‘hole, opening’ [initial]

PA	*taw-	‘open’ > *-ikaw-* ‘open’ (Berman 2006: 275, #65)
	*taw-	‘to open’ [A 2023]
	*taw-	‘hole, opening’ (Hewson 2024a: 277; cf. Hewson 1993: 192–193)
	*taw-	‘cut a hole’ (Hockett 1981: 69)
	*taw-	‘having an opening, leaving a gap’ [P 3091]
B	áá-wao-xkaa-ʔwa	‘it is crumbling down’ (Taylor 1969: 222)
	áak-áóo-hkaa-wa	‘it will fall away’ [FR 301]
	áak-ao-ks-kaa-wa	‘there will be a hole in it’ [FR 301] (medial *-ks-* ‘wood’)
	áak-aoo-ks-iisi-wa	‘she will have her plans fall through’ (lit. ‘have a hole in her boat’) [FR 301] (medial *-ks-* ‘wood’)
	ao-ks-tsí-t	‘gnaw a hole in it!’ [FR 301] (medial *-ks-* ‘wood’)
	ao-ni-htsí-t!	‘bite a hole through it!’ [FR 301]
	ao-níí-t!	‘pierce it!’ [FR 301]
	áak-aoo-ka-asi-wa	‘she will step on something and puncture her foot’ [FR 301]
	au'nikau	it has a hole in it (Tims 1889: 139)
	au'nitsĭmaie	he bites through it (Tims 1889: 112)
	áukskau	‘hole’ in *omí áukskau ómoχtaipstsàmmokoaiauaie* there was a hole, through which she (4 p.) peeped in at them repeatedly (Uhlenbeck 1938: 175)
	ikáy-ssp-omoo-sa!	part her hair! [FR 44]
	kaay-ínni-t	hold it open! [FR 44]

	kaaw-ahkó-yi	coulee/ditch/depression [FR 133]
	kaaw-áí’piksi-t!	open it! [FR 44]
	kaw-ohpáttsii-t!	break it open! [FR 44]
Ar	tóón-oti-’	/ton-oti-’/ ni/vii hole, opening; pothole; cave [L22697]
	ton-én-	/ton-en-/ vti make a hole in s.t. [L22539]
	tón-es-	/ton-es-/ vta pierce a hole in s.o. [L22541]
	ton-éx-	/ton-ex-/ vti pierce a hole in s.t. [L22545]
	ton-óh-	/ton-oh-/ vti make a hole in s.t. [L31019]
	ton-óót-	/ton-oot-/ vti make a hole using one’s teeth [L22588]
Ch	É-tovó’ko.	‘It’s chipped. (that is, has a gap or hole from being chipped).’
	É-tovó’-oo’e.	‘He is cracked. (for example, of a rock or mountain)’
	É-tovó’-án-a.	‘He took a chunk out of it.’
C	taw-â-w	‘it is open, it is an opening’
	taw-ah-am	‘s/he slashes s.t. open (as a path), s/he clears or marks s.t. (as a line)’
	taw-astê-w	‘it lies leaving a passage through’
	taw-in-am	‘s/he opens s.t. by hand’
M	taw-a·ʔtɛ-w	‘it has a gap in it, it gapes’ [H 3264]
	taw-a·h-am	‘he makes a hole in it by tool’ [H 3261]
	taw-ɛ·n-am	‘he tears a hole in it’ [H 3266]
O	daw-aa	‘there is a space, a gap, room’
	daw-abi	‘s/he makes room (for someone to sit)’
	daw-ishkan	‘make room for or in it (with foot or body)’
Mk	taw-a·w-i	‘it is an opening’ (Goddard 1994a: 163)
Sh	taw-âw-i	‘open, openness, space’ (Selstad 1998: 247)

	-taw-en(a)	‘to open (door)’ (Selstad 1998: 247)
Ps	tuw-alok-e	‘there is hole in it, it has hole in it’
	’-tuw-alok-ihkom-on	‘s/he puts hole through it by stepping on it’
	’-tuw-alok-ihtah-al	‘s/he puts hole through h/ by hitting h/’
	’-tuw-apotom-on	‘s/he looks through it (hole, window)’
	’ci-tuw-apu	‘s/he looks out or in from there’

The root *ao-* is not an “adjunct” in Frantz & Russell (2017). That means that it is not given its own entry in the dictionary. It only occurs in initial position as part of a stem. To find it then requires a pretty good understanding of the morphemic analysis of Blackfoot stems, and it is understandable why it might be overlooked at first. In contrast, the root *kaaw-* is listed as an “adjunct” under its own entry. It occurs in a preverbal position before the stem *ohpattsii-* TI ‘collide with’ in *kaw-ohpattsii-t* ‘break it open’ above, and otherwise in initial positions in the other examples.

The two Blackfoot reflexes are in complementary distribution, suggesting they stem from a single source. However, both involve irregular sound changes. First, the preconsonantal reflex *ao-* requires the spurious deletion of the initial PA *t and the spurious change of *aw > *ao*. Second, the prevocalic reflex *kaaw-* requires not only initial *t > *k*, but also a spurious vowel lengthening process. PA *a does not typically lengthen to long *a· in this environment, which can be seen by examining reflexes of PA *kaw- ‘down’. For example, *kaw-enam- TI ‘push down’ (with final *-enam- ‘by hand’) > B *kooni-* ∼ *-ikooni-*, as in *kooni-t!* ‘take it down (the tipi)!’ [FR 56] and *áaks-ikooni-ma* ‘she will take it down’ [FR 56]. Here, the sequence *awe > *oo* which does not shorten after an *i*-prefix. (See discussion in Berman 2006: 267 for the TA stem *kaw-en- TA ‘push down’ (with final *-en- ‘by hand’) > B *-ikoon-* TA ‘pull down’, though he only gives the non-initial stem in Blackfoot.) The vowel does not lengthen in this stem, even though the PA root is minimally different than PA *taw- ‘open’.

It is possible that there was contamination with another PA root, listed as *ta·w(aθ)- ‘gaping, open’ (Hewson 2024a: 269) or *ta·waθ- (NG) (Goddard 1980: 145). This might explain the long vowel in *kaaw-*, for example, but it would not explain the change of initial *t > *k*, the deletion of *t in *ao*, and the change of *a·w > *ao*. All in all, there are enough confounding factors for these reflexes that the the reconstruction seems rather uncertain to me.

6 Conclusion

This paper investigated three reconstructions that Berman (2006) argues involve a change of word-initial PA *t to B *k*. Such a change would be highly unusual, involving a major place change from coronal to dorsal. Ultimately, I found that PA *tepesk-at-wi* ‘it is night’ > B *ko'ko-wa* ‘it is night’ is well-supported by comparative evidence, the known relative chronology of sound changes in Blackfoot, and historical evidence. However, PA *tepeskw-i* ‘night’ > B *ko'ko-yi* ‘night’ is refuted by the established relative chronology of sound changes in Blackfoot. Finally, PA *taw- ‘hole, opening’ > B *kaaw- might* be correct, but would involve irregular sound changes. Even more interestingly, there appears to be a second Blackfoot reflex, *ao-* ‘hole, opening’, which is in complementary distribution with *kaaw-* but which also involves irregular sound changes. Since there are so many uncertain factors, it is not strong support for a *t > *k* change. Still, the fact that one of the three reconstructions is supported beyond a doubt means it would be worthwhile to investigate this claim further.

Finally, I reflect on the range of methods I used to examine the three reconstructions. These included comparative evidence, historical evidence from older written sources, and a careful consideration of the relative chronology of known sound changes. The cognate sets in particular provided rich and nuanced data regarding the morphophonology and semantic range of roots and stems. The semantic information was crucial in determining that *ao*- 'hole, opening' was a likely reflex of PA *taw-, and the phonological contexts were crucial for determining that *ao*- is in complementary distribution with *kaaw*- 'hole, opening'. Although most papers on Blackfoot historical phonology simply compare an abstract PA reconstruction to an abstract Blackfoot reflex, I recommend the use of cognate sets in future work. Regarding rule order, the description of sound changes in Blackfoot often implicitly includes rule order, as when Berman claims that "medial *a and medial *e become *i* but differ in morphophonemic treatment from inherited *i*" (Berman 2006: 266). However, I found it a useful exercise to formalize sound changes into rules and argue for their order.

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